

**KWAME NKRUMAH UNIVERSITY OF SCIENCE AND
TECHNOLOGY**

COLLEGE OF ENGINEERING

DEPARTMENT OF GEOLOGICAL ENGINEERING

INDUSTRIAL TRAINING REPORT

**REPORT ON INDUSTRIAL TRAINING AT
HYDROHOPE DRILLING AND CONSTRUCTION SERVICES**

COMPILED BY

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DECLARATION

I hereby declare that, except for quotations and information obtained from books, the internet, and other properly cited sources, this work is entirely the result of my own research. I further affirm that no part of this work has previously been submitted to any college or university in fulfillment of the requirements for any other degree.

Signature: _____

Date: _____

ABSTRACT

Internships provide students with the opportunity to apply classroom knowledge in real-world situations by integrating theoretical concepts with practical experience. This report presents a detailed account of the activities and experiences undertaken during my industrial training at Hydrohope Drilling and Construction Services.

ACKNOWLEDGEMENT

I am deeply grateful to God Almighty for granting me strength, guidance, and protection throughout the training period. I sincerely appreciate the management of Hydrohope Drilling and Construction Services for offering me the opportunity to undertake my industrial training with the organization. I also acknowledge the guidance and supervision of Mr. Ofosu Quainoo Isaac, whose support contributed greatly to the success of my training.

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CHAPTER ONE

INTRODUCTION

Date of visit

My Industrial Training at Hydrohope Drilling and Construction Services took place on December 7, 2025. The following day, an induction session was conducted, after which safety guidelines, workplace regulations, and the overall objectives of the visit were outlined. The internship program officially came to an end on January 10, 2026.

Organization of Report.

This report has four chapter. It includes an introductory chapter (chapter one), The literature review (chapter two), equipment and methodology (chapter three), and finally, conclusions and recommendations (chapter four).

CHAPTER TWO

LITERATURE REVIEW

Exploration Process

Desk Study

A desk study is conducted to collect as much information as possible about the project area. This involves obtaining geological maps, satellite images, and all available information related to water resources, not limited to groundwater alone. The long-term sustainability of the borehole and the potential impacts of climate change are also considered, rather than focusing solely on drilling.

Borehole Siting

Borehole siting involves identifying the most suitable location for drilling and estimating subsurface conditions. Although the exact underground conditions cannot be known without drilling, geophysical techniques—particularly electrical methods—are used to make informed estimates.

Difference Between Permeable and Impermeable Soil Types

Permeable soils, such as sand, allow water to pass through them, whereas impermeable soils, such as clay, restrict water movement. When a permeable layer holds water, it is referred to as an **aquifer**. Some aquifers may be confined beneath impermeable layers

and may be under pressure, causing water levels in boreholes to rise higher than in shallow wells. In rare cases, water may even flow out of the borehole without pumping.

Electrical Resistivity of Different Materials

Different geological materials possess different electrical resistivity values. Understanding these variations helps identify subsurface layers and locate potential aquifer zones using electrical resistivity methods.

Baseline Survey

Customer Preference

A competent borehole siter first engages the client and conducts a physical inspection of the area. The client may indicate preferred drilling locations, which are then evaluated for suitability.

Pollution Sources

Potential sources of contamination are identified, and unsuitable locations are excluded where necessary. Boreholes should be located at least **30 m** from latrines, septic tanks, and animal shelters, and at least **150 m** from refuse dumps and industrial facilities.

Visible and Oral Area Information

Information gathered from local residents includes:

- Existing water sources
- Seasonal drying of wells
- Water levels in nearby wells
- Previous drilling attempts
- Presence of perennial vegetation
- Occurrence of anthills
- Topographic variations

These observations assist in identifying suitable measurement locations.

How Electrical Sounding Works

Electrical current flows more easily through some materials than others. By inserting electrodes into the ground and applying electrical current, subsurface resistivity variations can be measured. Increasing the spacing between electrodes allows deeper penetration of the electrical current, enabling investigation of deeper geological layers.

CHAPTER THREE

EQUIPMENT AND METHODOLOGY

Equipment

- Surveyor's tape
- ADMT-200S-Y Water Detector

The ADMT-200S-Y water detector uses the Earth's natural electromagnetic field as its working source to study subsurface electrical structures for the delineation of aquifers for groundwater exploitation. The electrode and line measurement methods are used with this system, as shown in Figure 2 below.

Methodology

Traverse Line

A traverse line was established by laying a measuring tape along the preferred drilling locations.

Geophysical Survey (Groundwater Prospecting)

The ADMT-200S-Y water detector was used for groundwater prospecting. The electrodes were assembled, and the device and cables were properly connected. The two electrodes were then inserted into the ground along the traverse line at an initial interval (MN). One operator was responsible for the main unit and the M electrode, while an assistant handled the N electrode.

The midpoint between the two electrodes was measured, and data were recorded on the ADMT-200S-Y water detector at each measurement point. After each reading, the electrodes were moved equidistantly in the same direction until the entire profile measurement was completed.



Figure 1 Geophysical Survey

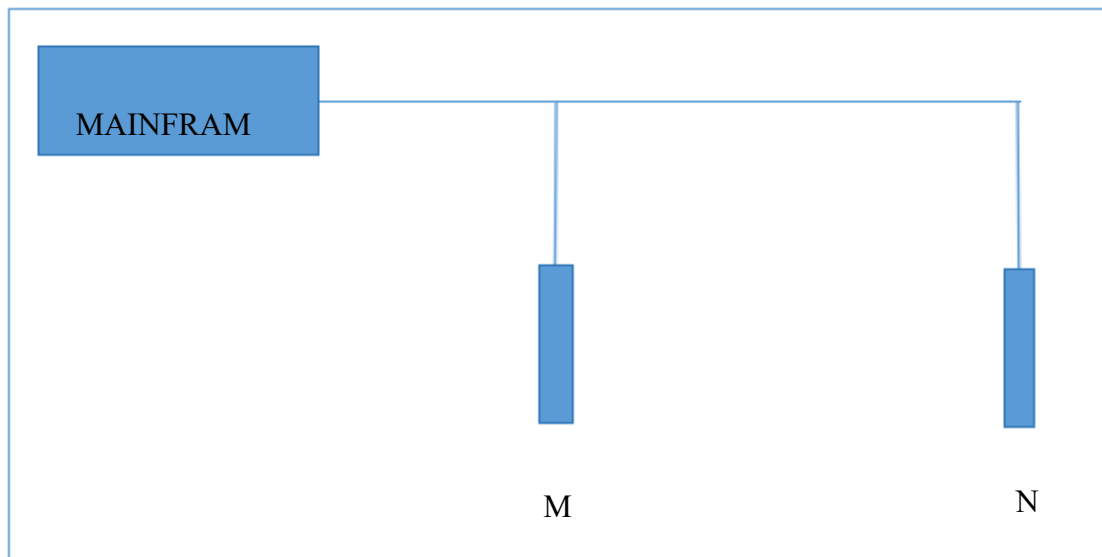


Figure 2 Electrode and line measurement methods

Interpretation of Data

The data captured by the instrument were analyzed using Aidu Intelligent Data Processing Software and presented as contour maps, allowing interpretations to be made regarding the location, extent, and estimated depth of geological anomalies.

The results are presented as contour maps in the appendix, where the horizontal axis represents the measurement point numbers and the vertical axis represents depth. Green regions indicate aquifer zones, while blue regions represent low-resistivity areas. Based on these results, three locations were identified and ranked as primary and alternative sites (first choice, second choice, and third choice). Site accessibility for drilling equipment was also considered during the selection and prioritization process.

Writing of Geophysical Survey Report

After the geophysical survey for groundwater prospecting and the interpretation of the data, a groundwater prospecting report was prepared. Proceeding to write or confirm the drilling specifications indicated that the groundwater prospecting exercise was successful. At this stage, the number of wells to be drilled and the required drilling depths were communicated to the drilling contractor.

Borehole Site Selection

The objective of the geophysical studies was to ensure a high success rate for the drilling works by applying the most appropriate and scientific approach to site selection. Furthermore, the studies ensured that boreholes were designed and constructed to provide year-round potable water to the beneficiary communities. The report on this has already been submitted.

Borehole Construction

General

Borehole construction took a few days after the geophysical investigations. A JCR DTH water well rig was used for the execution of the drilling works. Borehole Development

Borehole Construction Procedures

Drilling

To complete a borehole using 125 mm casing, a 300 mm hole was drilled using an insert bit with air or mud flush to the partially weathered rock. A 250 mm steel temporary surface casing was installed to hold back the overburden. A nominal 175 mm (7-inch) hole was then drilled through the underlying rock to the final depth using an air down-the-hole hammer button bit, sometimes with water injected into the air circulation during drilling.

When the hole finally intercepted the water-bearing aquifer, airlift discharges were measured at regular intervals during drilling. Drilling was stopped when sufficient water, in excess of 13.5 litres per minute, was discharged continuously by the well and the aquifer was adequately penetrated. Formation cuttings resulting from the drilling action were collected at one-metre intervals as drilling progressed. A driller's daily log provided a complete description of all formations encountered, metres drilled, number of hours spent, and other relevant details.



Figure 3 A 250 mm steel temporary surface casing



Figure 4 A JCR DTH water well rig



Figure 5 Borehole logging

Borehole Design

Plain and slotted PVC pipe casings were used to line the borehole. They had a diameter of 125 mm and were procured from Interplast Company, registered in Ghana.

All PVC pipes were threaded, whereas the slotted pipes, also known as screens, had slot intervals measuring up to 6 mm and an average opening width of about 1.2 mm.

Two basic borehole designs are known; in both cases, the PVC lining was installed to the full depth of the borehole. In the more common design, the screen position started close to the bottom of the hole, spanning the water-bearing zone. The other design type was used in exceptional cases where it became necessary to use screens separated by plain casing. In this case, the lower screen was placed in the lower part of the hole to tap water flows from the basement rock, while the upper screen was placed against the gritty water-bearing levels found in the weathered zone overlying the basement rock. The length of the screen depended on the thickness of the water-bearing zone(s).

The bottom of the PVC screen was fitted with a 1 m length of plain PVC to serve as a sump for fines that could not be removed during borehole development. The lower end of the plain PVC was sealed in concrete, and the whole unit was referred to as a bail plug.

Formation Stabilizer, Backfill, and Surface Grout

After the placement of the PVC screens and plain casings, a formation stabilizer was placed in the annular space between the screens and the borehole wall. The formation stabilizer, consisting of a graded gravel pack (quartz grains), was placed to a minimum of 5 m above the top of the upper screen. The one-metre column of the annular space immediately above the gravel stabilizer/filter media was filled with a packaged clay seal or cement grout to prevent clayey and silty water from the overburden draining directly into the gravel pack.

The remaining annular space above the clay seal was backfilled with formation cuttings to within 4 m of the surface, or 10 m at locations with doubtful sanitary conditions. After borehole development, the upper 4 m of the annular space was then filled with cement grout to prevent surface water contamination of the borehole.

Borehole Development

After the installation of the PVC lining and the filter media, the borehole was developed to remove formation cuttings and other materials from the water-bearing zones, as well as fine material from the filter media.

Development was carried out using the air-flush method. The air-flushing process was repeated several times until the water in the borehole was judged to be free of sand and micaceous material. Borehole development was carried out by the drill rig crew.

Borehole Capping

All borehole lining was completed with the PVC casing extending to about 0.8 m above ground level. The borehole was then temporarily capped with a piece of PVC purposefully designed to protect the borehole until the pumps were installed. Borehole Yield Testing.

Borehole Yield Testing

Pumping Equipment and Water Level Measurements

A pumping test was conducted on each successful borehole drilled. For this purpose, the contractor mobilized to the site a mobile unit made up of electric submersible pumps and accessories, manned by a pumping test crew. The pumping assembly was fitted with a throttling device (to regulate discharge), complete with a suitable power source, controls, and other appurtenant structures.

The pumping rate during the tests was determined by measuring the time taken to fill a bowl of known volume. Water level measurements were taken from a convenient reference point—the top of the PVC borehole lining. Results of borehole tests were reported by completing appropriate forms designed for the purpose.

The contractor conducted a pumping test on each borehole constructed for a period of 24 hours at a constant rate. The pumping rate was consistent with the capacity of the borehole.

CHAPTER FOUR

CONCLUSIONS AND RECOMMENDATIONS

Conclusions

The industrial training broadened my understanding of engineering practices and enabled me to apply classroom knowledge to real-world field operations.

Recommendations

- Boreholes should not exceed **100 m** depth unless technically justified.
- Drilling should be supervised by a qualified hydrogeologist.
- Boreholes must be drilled at designated locations and depths.
- Proper cement grouting should be applied to prevent contamination.
- Pumping tests should be conducted to determine sustainable yield.

APPENDIX